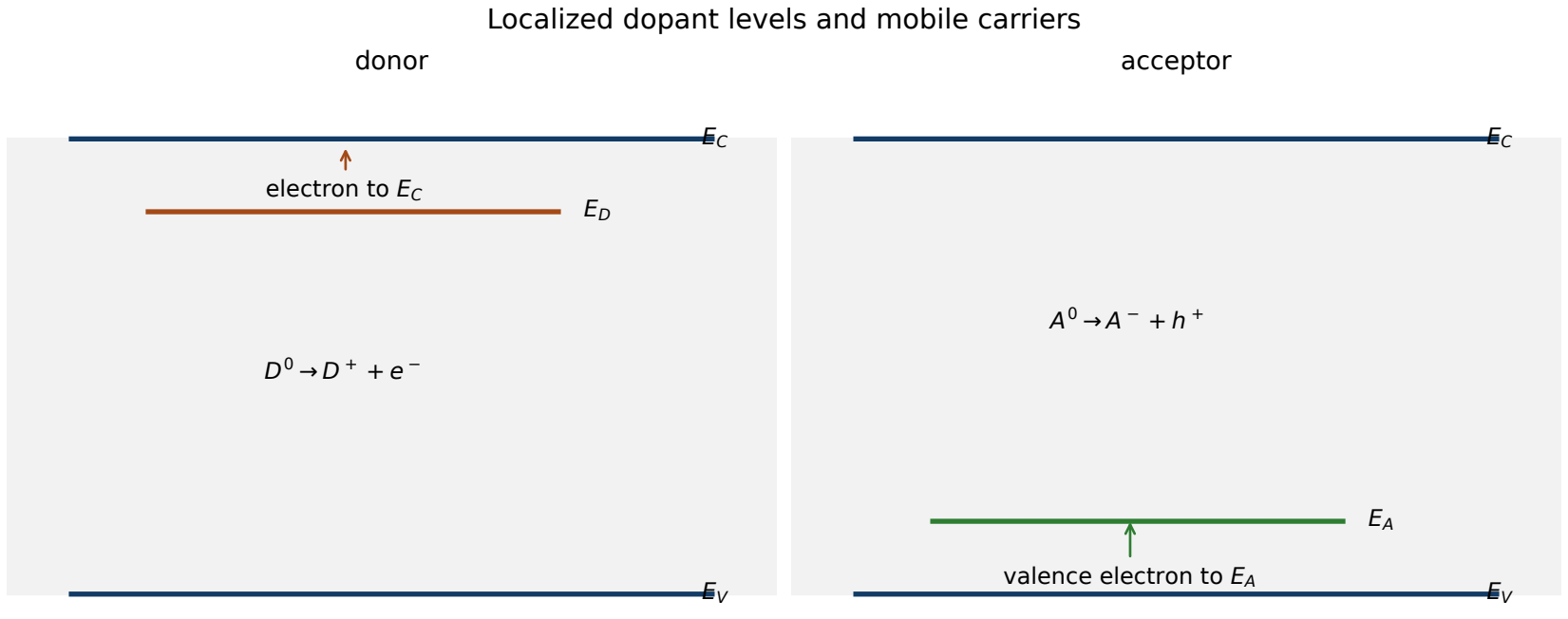
Localized dopant levels and mobile carriers
donor
acceptor
E_C
electron to E_C
E_D
$D^0 \rightarrow D^+ + e^-$
E_V
E_C
$A^0 \rightarrow A^- + h^+$
E_A
valence electron to E_A
E_V